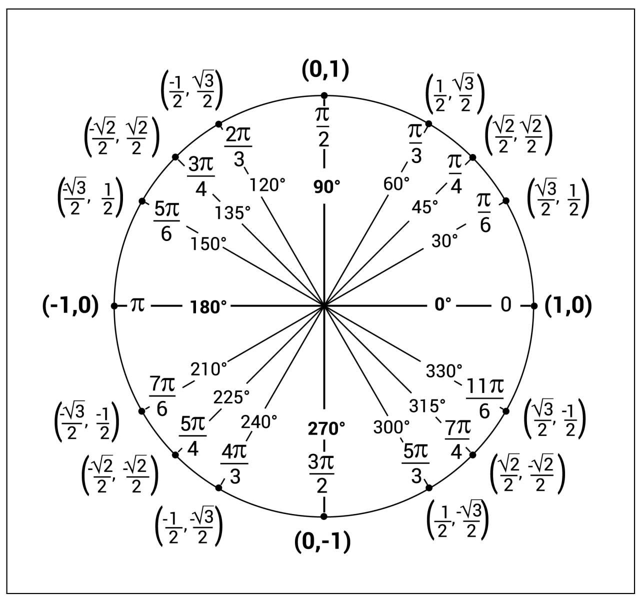
Math 125

Extra Unit C practice 2

The problems that follow are examples of questions I've asked on assessments in past semesters. They are meant to give you a sense of the style of problems on assessments, but not necessarily the content of the problems. For example, just because a particular concept doesn't appear in these practice problems doesn't mean that you aren't responsible for knowing it for our assessments. You're responsible for knowing anything in the notes, the past final worksheets, and the Brightspace problems. The formulas below will be provided on Opportunity C.

Formulas

- Integral formulas
  - $\int cf(x)dx = c\int f(x)dx$
  - $\int [f(x) \pm g(x)]dx = \int f(x)dx \pm \int g(x)dx$
  - $\int kdx = kx + C$
  - $\int x^n dx = \frac{x^{n+1}}{n+1} + C \ (n \neq 1)$
  - $\int \sin(x)dx = -\cos(x) + C$
  - $\int \cos(x)dx = \sin(x) + C$
  - $\int \sec^2(x)dx = \tan(x) + C$
  - $\int \csc^2(x)dx = -\cot(x) + C$
  - $\int \sec(x)\tan(x)dx = \sec(x) + C$
  - $\int \csc(x)\cot(x)dx = -\csc(x) + C$
  - $\int e^x dx = e^x + C$
  - $\int \frac{1}{x}dx = \ln|x| + C$
  - $\int f(g(x))g'(x)dx = f(g(x)) + C$
- $\left(f^{-1}\right)'(x) = \frac{1}{f'(f^{-1}(x))}$
- Fundamental Theorem of Calculus
  - $\int_a^b f(x)dx = F(b) - F(a)$
  - $\frac{d}{dx}\left(\int_a^x f(t)dt\right) = f(x)$
- Law of natural growth/decay
  - $Q'(t) = kQ(t)$
  - $Q(t) = Q_0 e^{kt}$
  - $T'(t) = k(T(t) - T_s)$



C1 - **The integral**

(a) If $\int_{-5}^{-3} f(x)dx = 2$ and $\int_{-5}^{-3} [-2f(x) - 3g(x)]\,dx = 23$, find $\int_{-3}^{-5} g(x)dx$.

$$23 = -2\int_{-5}^{-3} f(x)dx - 3\int_{-5}^{-3} g(x)dx = -2\cdot 2 + 3\int_{-3}^{-5} g(x)dx$$

$+4 \qquad +4$

$$27 = 3\int_{-3}^{-5} g(x)dx \rightarrow \boxed{\int_{-3}^{-5} g(x)\,dx = 9}$$

(b) A race car accelerates from rest to try to travel 1,000 feet to the finish line in 4 seconds. The velocity of the car, in feet per second, is recorded at each half of a second except at 2 and 2.5 seconds, as shown in the table below. In answering the following questions, you should assume that velocity is increasing.

time	0	0.5	1	1.5	2	2.5	3	3.5	4
velocity	0	30	80	140	x	y	440	480	500

(i) Find values for x and y that would definitively show the car did not cross the finish line within 4 seconds. Your answer should demonstrate how you know these values of x and y work.

Choose low values for x and y: $x = 150$, $y = 160$

right-hand sum is overestimate: $0.5\cdot(30+80+140+150+160+440+480+500)$

$0.5\cdot(1980) = 990 < 1000$ so car does not travel 1000 ft.

(ii) Find values for x and y that would definitively show the car did cross the finish line within 4 seconds. Your answer should demonstrate how you know these values of x and y work.

Choose high values for x and y: $x = 420$, $y = 430$

left-hand sum is underestimate: $0.5\cdot(0+30+80+140+420+430+440+480)$

$0.5\cdot(2020) = 1010$ so car does travel at least 1000 ft

C2 - The Fundamental Theorem of Calculus

(a) Find the interval where the function

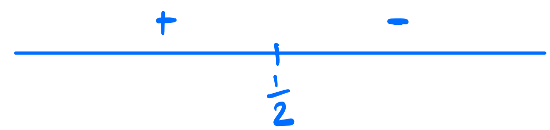
$$g(x) = \int_0^x \frac{dt}{t^2 - t + 1}$$

is concave up.

$g'(x) = \frac{1}{x^2 - x + 1}$

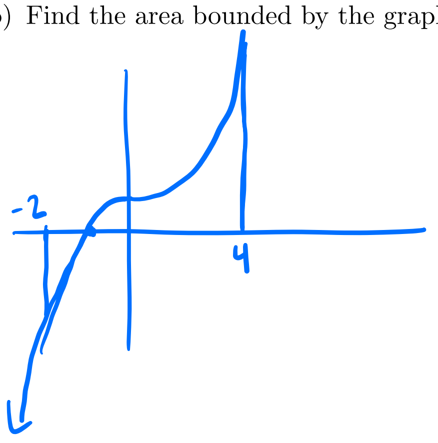
$g''(x) = \frac{-(2x-1)}{(x^2-x+1)^2} = 0$ @ $x = \frac{1}{2}$

$\leftarrow$ DNE nowhere



$\boxed{(-\infty, \frac{1}{2})}$

(b) Find the area bounded by the graph of $y = x^3 + 1$, the x-axis, and the lines $x = -2$ and $x = 4$.



$\int_{-2}^{-1} (x^3+1)\,dx = \frac{1}{4}x^4 + x \Big|_{-2}^{-1} = \frac{1}{4} - 1 - (4-2) = -2.75$

$\int_{-1}^{4} (x^3+1)\,dx = \frac{1}{4}x^4 + x \Big|_{-1}^{4} = 64 + 4 - (\frac{1}{4} - 1) = 68.75$

$68.75 + 2.75 = \boxed{71.5}$

C3 - **Integration by substitution**

Evaluate the following integrals.

(a) $\displaystyle\int \frac{4x+2}{(x^2+x)\sqrt{\ln(x^2+x)}}dx$

$u = x^2+x$

$du = (2x+1)\,dx$

$dx = \frac{du}{2x+1}$

$\int \frac{2(2x+1)}{u\sqrt{\ln(u)}} \cdot \frac{du}{2x+1} = \int \frac{2}{u\sqrt{\ln(u)}}du$

$w = \ln(u)$

$dw = \frac{1}{u}du$

$du = u\,dw$

$= \int \frac{2}{u\sqrt{w}} \cdot u\,dw = \int 2w^{-\frac{1}{2}}dw = 4w^{\frac{1}{2}} + C$

$= 4\cdot(\ln(u))^{\frac{1}{2}} + C$

$\boxed{= 4\cdot(\ln(x^2+x))^{\frac{1}{2}} + C}$

(b) $\displaystyle\int_0^{\frac{\pi}{4}} \sec^2(x)\left(1+\tan^2(x)\right)^2 dx$

$u = \tan x$

$du = \sec^2 x\,dx$

$dx = \frac{du}{\sec^2 x}$

$u(0) = \tan(0) = 0$

$u(\frac{\pi}{4}) = \tan(\frac{\pi}{4}) = 1$

$\int_0^1 \cancel{\sec^2(x)} \cdot (1+u^2)^2 \cdot \frac{du}{\cancel{\sec^2(x)}} = \int_0^1 (1+2u^2+u^4)\,du = u + \frac{2}{3}u^3 + \frac{1}{5}u^5 \Big|_0^1$

$= \boxed{1 + \frac{2}{3} + \frac{1}{5}}$

C4 - Inverse functions and their derivatives

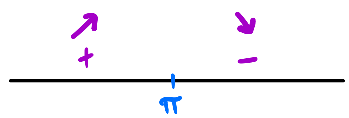
(a) For each function below, determine if it is one-to-one.

(i) $h(x) = x^7 + x^3 + x - 100$

$h'(x) = 7x^6 + 3x^2 + 1 > 0 \rightarrow$ h is one-to-one

(ii) $g(x) = \displaystyle\int_{\pi}^{x} \frac{\sin(t)}{t}\, dt$

$g'(x) = \dfrac{\sin(x)}{x} = 0$ @ $x = \pi$ (among others)



$\rightarrow$ g(x) is not one-to-one

(b) Let $f(x)$ be defined as

$$f(x) = x^3 + 3\sin(x) + 2\cos(x)$$

Find $[f^{-1}]'(2)$.

$f(0) = 0^3 + 3\sin(0) + 2\cos(0) = 2 \rightarrow f^{-1}(2) = 0$

$f'(x) = 3x^2 + 3\cos(x) - 2\sin(x)$

$f'(0) = 3$

$$(f^{-1})'(2) = \frac{1}{f'(f^{-1}(2))} = \frac{1}{f'(0)} = \boxed{\frac{1}{3}}$$

C5 - Exponential growth and decay

(a) If Cobalt-60 has a half-life of 5.24 years, how long will it take for a 100 mg sample to decay to 1 mg? Round your answer to the nearest tenth of a year.

$$\frac{1}{2} = 1 \cdot e^{k \cdot 5.24}$$

$$\ln\left(\frac{1}{2}\right) = k \cdot 5.24$$

$$k = \frac{1}{5.24} \cdot \ln\left(\frac{1}{2}\right)$$

$$Q(t) = 100\, e^{kt}$$

$$1 = 100\, e^{kt}$$

$$\frac{1}{100} = e^{kt}$$

$$\ln\left(\frac{1}{100}\right) = kt$$

$$t = \frac{\ln\left(\frac{1}{100}\right)}{k} = \boxed{\frac{\ln\left(\frac{1}{100}\right)}{\frac{1}{5.24}\ln\left(\frac{1}{2}\right)}}$$

(b) An investment account grows at an annual rate of 4%, compounded continuously.

(i) What is the value of the account at the moment it is growing at a rate of \$10,000 per year?

$$Q'(t) = k\,Q(t)$$

$$10000 = 0.04 \cdot Q(t)$$

$$\rightarrow \quad Q(t) = \frac{10000}{0.04} = \boxed{250{,}000}$$

(ii) How fast is the account growing at the moment it is worth \$1,000,000?

$$Q'(t) = 0.04 \cdot 1000000$$

$$= 40{,}000 \ \$/\text{year}$$